

#facebookdown: Time to panic or detox? Understanding users' reactions to social media outage

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ABSTRACT

Non-use, particularly involuntary non-use, is an under-researched topic in HCI research, even though it has become quite common nowadays due to frequent digital outages. How do users react to social media outage? Do they become anxious? Or, do they enjoy these brief episodes of social media detox? To answer these questions, we conducted a topic modeling analysis of 223,815 tweets that used the hashtag #facebookdown during the major Facebook outage on 10/4/2021. We uncovered 10 major themes of users' reactions towards social media outage. Results showed that most users complained, mocked and showed desperation about the outage situation, and during the outage period, increased their quest for other social-media alternatives. Also, surprisingly, many users celebrated the detox from Facebook rather than wishing it to come back as soon as possible. Results offer design implications for practitioners who would like to better respond to future outages.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Empirical studies in HCI**;

KEYWORDS

Social media outage, involuntary non-use, Topic Modeling

ACM Reference Format:

Mengqi Liao and S. Shyam Sundar. 2022. #facebookdown: Time to panic or detox? Understanding users' reactions to social media outage. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3491101.3519674>

1 INTRODUCTION

Facebook and its family of apps (i.e., Messenger, Instagram, and WhatsApp) together have the largest user base of any social media platform in the world. Over 3.5 billion people worldwide rely on these apps to communicate with family and friends, distributing messages, and conducting daily business operations. Therefore, when users realized that they could not use Facebook in early October 2021, they got confused, annoyed, and even outraged. Soon

after Facebook's outage, Twitter was flooded with Facebook users who complained about this incident. The hashtag #facebookdown started trending, with users expressing their thoughts towards Facebook's outage, thereby providing valuable data for studying their immediate reactions towards social media outage. Some users on Twitter expressed their confusion and asked for assurances from other users, some complained about the company and Mark Zuckerberg, and others made fun of the situation or expressed their appreciation of this short social-media detox.

In HCI research, existing literature has examined voluntary social media non-users, but how users react to involuntary non-use is still under-researched. In the current study, we scraped all the English language tweets from 10/4/2021 (15:30 UTC)-10/6/2021 (11:59 UTC) that used the hashtag #facebookdown during the recent Facebook outage by utilizing Twitter APIv2, which returned 223,815 tweets in total. By conducting topic modeling using LDA (Latent Dirichlet Allocation), common themes underlying users' discussion of the involuntary non-use of information communication technology were uncovered. These results can offer design implications for practitioners who are interested in being better prepared for future outages.

2 RELATED WORK

Lately, researchers have called for more research on non-users, given that majority of HCI research is primarily about users [3, 15]. According to Wyatt [19], non-use can be either voluntary (i.e., the rejectors) or involuntary (i.e., the excluded). Extant research on the causes and motivations for voluntary non-use of social media suggests that users are concerned about surveillance, mood, the availability of other social media platforms, intention for self-control, disinterest, and disenchantment or frustration [4, 9, 11]. According to Baumer et al. [4], one of the biggest differences between voluntary and involuntary non-use is the absence of user agency with the latter. Therefore, we would expect to see more frustration among involuntary non-users. However, this frustration among non-users is seldom studied by past researchers, except a few who focus on a specific group of users, such as those with certain physical disabilities [15]. Yet, the experience of involuntary non-use can be quite universal and common among general users as well. In fact, according to Downtdetector (<https://downtdetector.com/insights/>), in 2020 alone, there were 12 major social media outages worldwide across many social platforms (i.e., the number of user reports of outage is more than 3,000). Thus, understanding users' concerns and reactions towards involuntary non-use becomes pressing and important, as it helps us better prepare for the not-so-uncommon outage incidents and restore users' trust in the long run.

A recent qualitative study gauged this issue (i.e., the involuntary loss of use) by adopting a future-oriented scenario-based approach

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CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519674>

and let participants imagine how internet outage will impact their life and society if it happens in the future [9]. They found that users tended to have mixed reactions towards internet outage, as some participants think that it can help improve the quality of their social and personal relationships, their psychological well-being, their cognitive abilities without reliance on the internet, and their productivity, while others think it would reduce their connectivity with others, hinder their information access, their emotional health (mostly due to panic and anxiety) and their quality of life in general, as internet has become so indispensable in their lives. Similar to the penetration of the internet, social media has also become so pervasive now, missing which will cause major disruptions in the daily lives of millions of individuals. Instead of taking a future-oriented scenario-based approach, with the recent occurrence of the Facebook outage event and the help of Twitter data, we look into users' real reactions towards ICT's outage at a much larger scale as they learn by answering the following research question:

RQ1: What are the most common user reactions when facing a major social media outage?

Despite some users' anticipation that staying away from social media might be beneficial for their quality of life, the majority of users might still feel frustrated by involuntary nonuse. According to Uses & Gratifications Theory, individuals' motivations guide their usage of media in order to fulfill their needs [10]. In the context of social media, many common motivations guide usage, including social interaction, entertainment, passing time, information seeking, information sharing, convenience, companionship and altruism [1, 5, 12, 14, 16, 18]. Park [13] investigated how Korean smartphone users reacted to a six-hour communication blackout, by conducting in-depth-interviews with 70 participants, and discovered two types of dependence on communication technologies—functional dependence (relating to how mobile phones serve an important instrumental role), and existential dependence (the extent to which one is deeply attached or addicted to mobile phones). Given such dependency, frustration will likely dominate users' reactions towards any major unplanned outage of communications. Therefore, we propose the following hypothesis:

H1: Users' reactions towards social media outage will be dominated by frustration.

3 METHOD

Using Twitter API v2, we scraped every tweet using the hashtag #facebookdown (and its variations: #Facebookdown, #facebook-Down, #FacebookDown)¹ from 10/4/2021 (15:30 UTC)-10/6/2021 (11:59 UTC) in English. We scraped the original tweets to make sure we captured the complete text of each tweet. After removing duplicate tweets, the final corpus included a total of 223,815 tweets. Upon preprocessing the data, we created a document-term matrix (DTM) with unigrams (i.e. n-gram=1)² (See Appendix A1 for details of data preprocessing in which 23 self-identified stopwords and the default stopwords from quanteda package were removed).

¹The #facebookdown hashtag and its variations were chosen because 1) majority of the users used it to communicate Facebook outage, making it the number one trending topic on Twitter, 2) users also used other hashtags alongside #facebookdown (e.g. #instagramdown, #whatsappdown, #messengerdown), allowing me to capture users' responses towards the outage of Facebook's family of apps at the same time

²Materials needed for data replication can be found here: <https://osf.io/b2qn4/>

3.1 Topic Modeling

Using the DTM, we generated a topic model with the LDA approach and Gibbs sampling method by adopting the topicmodels package in R. Although LDA is known to be better when dealing with large datasets, which is the primary reason why we chose this approach to conduct the topic modeling, it involves subjective judgments when deciding the number of topics to generate [21]. That is, we need to specify the desired number of topics (i.e., k) to build the model. A 10-topic model was retained based on both statistical and theoretical considerations. Statistically, the 10-topic model generated one of the best model fits based on three different matrices identified by past researchers [2, 6, 7]. Theoretically, the 10-topic model also generated the most interpretable results among the potential models (see Appendix A2 for more details about the model selection process). The per-tweet-per-topic probabilities were calculated using the tidytext package in R. Each tweet has different probabilities of covering each of the 10 topics. Therefore, each tweet has 10 different probabilities to be classified as the 10 different topics respectively. We selected the dominant topic of each tweet, to represent the theme of the tweet, by using the slice_max function. Using this method, we could calculate the number of tweets that covered each of the 10 topics as well as the frequency of the 10 topics across the corpus³. In order to understand how users' reactions change during the 7-hour outage, we also created a graph to gauge the trend of each topic's proportion between 10/4/2021, 15:30 (UTC) to 10/5/2021, 00:00 (UTC) using polynomial interpolation method which helps us create a smooth trend graph. In order to interpret the theme of each topic, in addition of making interpretation based on the list of top terms, we also randomly selected 1,000 tweets from the dataset, and then chose the most representative quotes that contained the top 20 terms for that corresponding topic (presented in the results section) to help us better understand how these terms were used in the real tweets.

4 RESULTS

4.1 Results from the LDA model

Before the LDA model, we generated word cloud among all the common words that are used by users when discussing the Facebook outage incident (see Appendix C). In general, terms like "server", "now", "disabled", "mark zuckerberg", "outage", "back", "delete" were used most often by users. Figure 1 showed the output from the 10-topics model, as well as the top 20 terms with the highest frequency within each topic.

Based on the results from the topic model, Table 1 shows the percentage of tweets about each of the ten topics. In general, the overall negative reactions towards the outage (i.e., topic 7, 2, 8, 4, 1 and 5) appears to be higher in volume than tweets that are celebratory of the outage (6 and 10), thereby supporting H1.

Figure 2 shows the trend of each topic's proportion during the outage. We can observe from the graph that in general, most user reactions did not vary much across time. At the beginning of the outage, Topic 2 (desperate effort to log in) dominated the twitterverse, but as time went by, the only topic that increased significantly

³Since some tweets have equal possibilities to be classified as more than one topic ($N=109,009$), adding up the number of tweets within each topic would result in a number that is higher than the total number of tweets in the corpus.

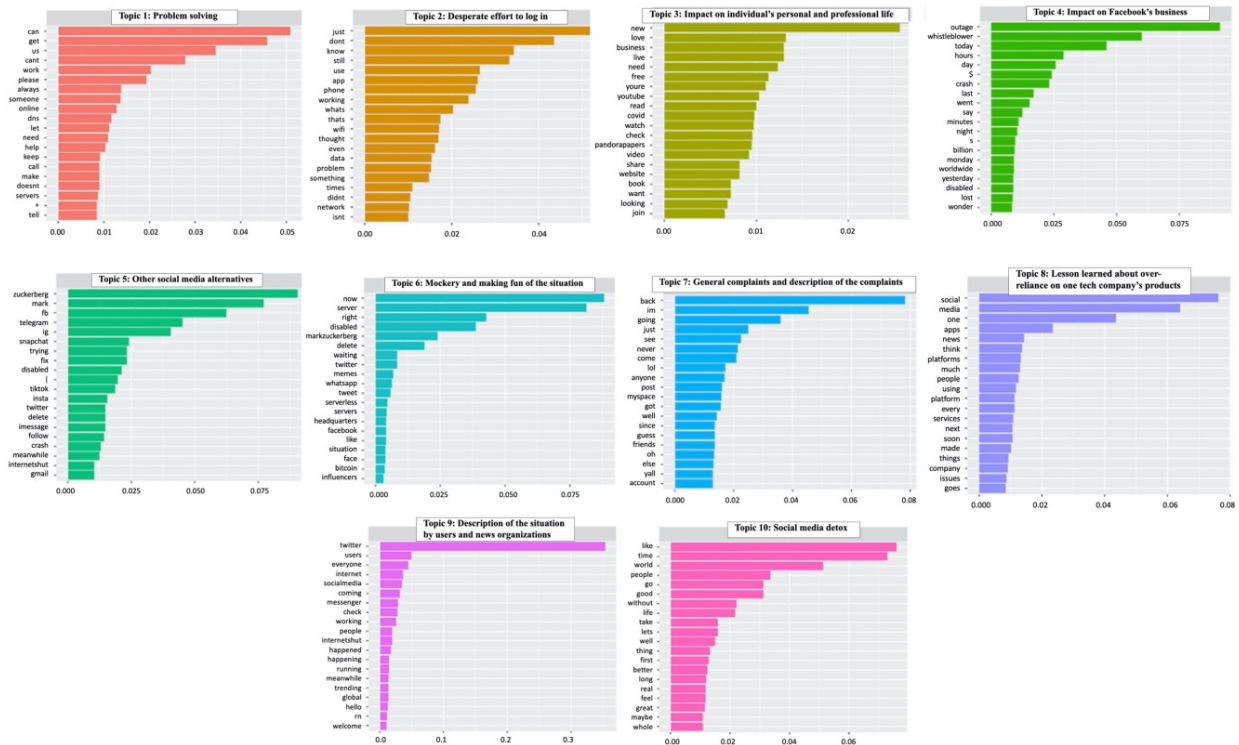


Figure 1: Top 20 terms within each topic (based on the per-topic-per-word probabilities, beta) for the 10-topic LDA model

Table 1: Percentage of tweets under each topic

Topic	Theme	Percentage of tweets
7	General complaints and description of the complaints	N= 38,455 (11.55%)
6	Mockery and making fun of the situation	N= 37,288 (11.20%)
9	Descriptions of the situation by users and news organizations	N= 36,621 (11.00%)
10	Social media detox	N= 35,371 (10.63%)
2	Desperate efforts to log in	N= 32,826 (9.86%)
8	Lesson learned about over-reliance on one tech company's products	N= 31,528 (9.47%)
4	Impact on Facebook's business	N= 31,211 (9.38%)
1	Problem solving	N= 30,455 (9.15%)
3	Impact on individuals' personal and professional life	N= 29,793 (8.95%)
5	Other social media alternatives	N= 29,276 (8.80%)

in its proportion is Topic 5, which is users searching for other social media alternatives, while the general descriptions of the outage by users and news organizations (Topic 9) went down in number.

4.2 Interpretation of the theme of each topic based on representative quotes

4.2.1 Topic 1. Problem solving. Some users/companies took advantage of the Facebook outage to promote their services that were claimed to be able to help with this situation, e.g., Tweet 43498: “#facebookdown Cloudflare senior vice president Dane Knecht notes that Facebook's border gateway protocol routes, BGP helps networks

pick the best path to deliver internet traffic” (gamma⁴ of topic 1, .265). Others provide useful information to other users in the hope of solving the problem, e.g. Tweet 132070: “Tech talk.....Facebook etc is down due to DNS records being removed from the global routing tables. When these are removed Web addresses cannot resolve to an IP #facebookdown” (gamma of topic 1, .262)

4.2.2 Topic 2. Desperate efforts to log in. Many users shared they desperately wanted to log in when facing the outage, e.g. Tweet 90161: “Today I really looked stupid.. I did restart my phone several times, I did airplane and network several times, Connected

⁴Gamma represents the estimated proportion of words from that tweet that are generated from a specific topic

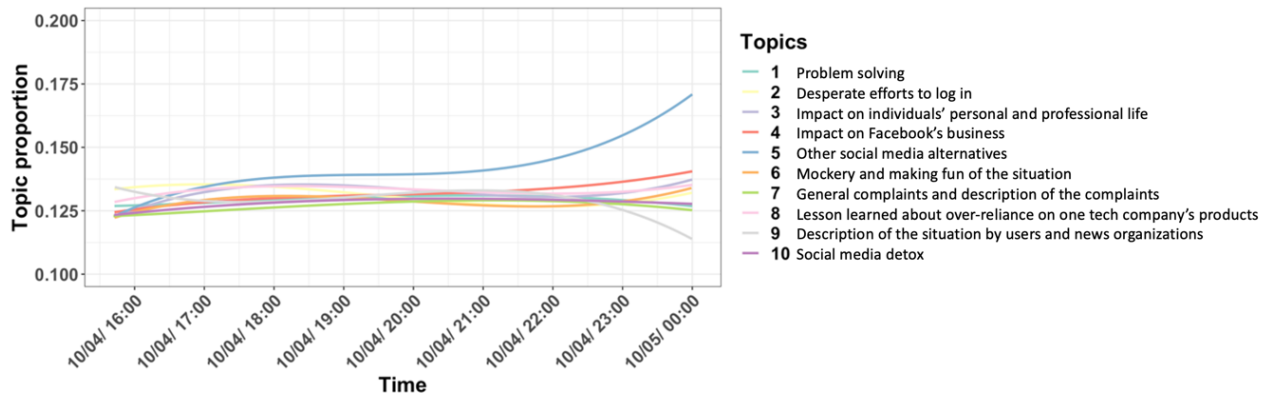


Figure 2: Over-time trend in each topic's proportion during the outage between 10/4/2021, 15:30 (UTC) to 10/5/2021, 00:00 (UTC)

to 3 different wifi, Even asked Google whats wrong with my data settings and wifi settings #instagramdown #facebookdown #WhatsAppDown" (gamma of topic 2, .338); Tweet 54617: "After 3 times Restarting my Phone, 5 times Switch On/Off my Wifi Device, 10 times On/Off My Data, And After a long Blaming on Phone Company When I Come on Twitter #facebookdown" (gamma of topic 2, .265).

4.2.3 Topic 3. Impact on individuals' personal and professional life. Users generally shared their concerns or experiences of how the Facebook outage impacted their life, e.g., Tweet 38974: "With a local event we are putting on and trying to promote it. Plus trying to promote our podcast and youtube channel. Facebook being down is not making things easier. (link to podcast in bio for those interested) #facebookdown #podcastlife" (gamma of topic 3:.265); Tweet 187174: "Infinite loop click FB, click IG, click Whatsapp, click FB, click Whatsapp, click FB #facebookdown #instagramdown #WhatsAppDown" (gamma of topic 3: .180).

4.2.4 Topic 4. Impact on Facebook's business. Many of this group's tweets expressed their opinions about how the outage impacted the business of Facebook, e.g., Tweet 208285: "Tweet 10hrs of #facebookdown time, what does this mean in \$\$\$? \$12m every minute for 10hrs straight. Mark Zuckerberg lost \$7.2b in a day due to outage. I call it The MARKOCALYPSE #whatsappdown #instagramdown #InternetShutDown" (gamma of topic 4, .333).

4.2.5 Topic 5. Other social media alternatives. Users also shared how they were using other social media alternatives to pass the time during Facebook and its family apps' outage, e.g. Tweet 51126: "As whatsapp has shutdowns let not get bored <https://t.co/v1cyD4meXq> #Whatsapp #facebookdown #Telegram #imessage #snapchat #DeleteFacebook" (gamma of topic 5, .265), while others expressed how the other social media platforms were not meeting their needs, e.g. Tweet 146050: "There are only #Snapchat #Twitter for Android left because the #Telegram #Tiktok were saturated on the most annoying Monday #whatsappdown #facebookdown #instagramdown" (gamma of topic 5, .138).

4.2.6 Topic 6. Mockery and making fun of the situation. Many users also flooded to Twitter to make fun of the situation, e.g., Tweet 185474: "haha, I am finding everyone here, literally! #facebookdown

#WhatsAppDown #instagramdown #Snapchat #tiktokdown #InternetShutDown" (gamma of topic 6, .158); Tweet 50798: "Screw them all, let's move back to MSN lol #facebookdown #whatsappdown #DeleteFacebook" (gamma of topic 6, .148)

4.2.7 Topic 7. General complaints and description of the complaints. Several users complained about the situation, e.g., Tweet 154729: "WTF is happening to the social giants!? #facebookdown" (gamma of topic 7:.156). Some users were also describing the enormity of user reactions to the incident, e.g., Tweet153954: "Everyone going nuts over #facebookdown #instagramdown and #whatsappdown as if the apocalypse is happening. #myspace : FRIEND 4 FRIEND ANYONE?" (gamma of topic 7, .209).

4.2.8 Topic 8. Lesson learned about over-reliance on one tech company's products. Many users also reflected on how social media like Facebook were so important and indispensable in their everyday lives, e.g., Tweet 180749: "I guess now is the perfect example to why we have to rely on another messaging platform other than Whatsapp... It makes no sense to totally rely on ONE SINGLE COMPANY for the majority of social media... #facebookdown" (gamma of topic 8, .265); Tweet 151360: "A moment of silence for everybody that are unable to login to other websites they are using by relying on their Facebook account to login... #facebookdown #DeleteFacebook"(gamma of topic 8, .150).

4.2.9 Topic 9. Description of the situation by users and news organizations. Many tweets did not express a specific opinion, but just offered a description of the current outage situation, e.g., Tweet 49881: "Information source Note: Facebook, WhatsApp, Instagram and Messenger are currently experiencing outages in multiple countries; incident not related to country-level internet disruptions or filtering #FacebookDown" (gamma of topic 9, .262), Tweet 45316: "Whatsapp, facebook and instagram users arriving at twitter just to check if it's down #whatsappdown #facebookdown" (gamma of topic 9, .175).

4.2.10 Topic 10. Social media detox. Some users also thought this outage was a good thing and that Facebook should remain down for people to experience a complete social media detox, e.g., Tweet 155129: "Facebook being down is healthier for all humans. Maybe this is the time we reunite with ourselves, talk to our soul, listen to our voice and feel peace. This is the detox we need. #facebookdown

#life #LifeLessons #detox #Peace" (gamma of topic 10, .262), Tweet 90961: *"Well...then- the world pauses. Facebook and Instagram go mysteriously offline and for one moment- the world becomes quiet of the chaos. Breathe a bit lighter. #facebookdown #igdown"* (gamma of topic 10, .254).

5 DISCUSSION

Results showed that users expressed a wide range of opinions regarding the outage. While the majority of users were complaining about it, some were making fun of the situation and commiserating with other Facebook users on Twitter. The three most common topics were user complaints (topic 7), mockery (topic 6), and descriptions of the situation by users and news organizations (topic 9). Other than the general reactions, surprisingly, more users expressed their appreciation of detox from Facebook (topic 10) than their frustration of how the outage impacted their personal and professional lives (topic 3). Yet, during the time when the outage happened, we could see an upward trend in the number of users searching for other social media alternatives (topic 5). This finding, coupled with the fact that users are using another social media platform (i.e., Twitter) to communicate how good it is to experience Facebook detox, suggests that users may be experiencing cognitive dissonance [8]. This is especially likely given that a very popular user reaction was desperation when the outage happened (topic 2).

These results have practical implications for designers who would like to better prepare for future outage and restore users' trust. Grandhi et al. [9] found that many users mentioned their concerns and potential anxiety of internet outage. Similarly, we found a considerable number of users shared their experience of how the outage had a negative effect on their personal life, and how they seek out other social media alternatives to address their concerns, even though some of them were dissatisfied with the alternatives. There were also some users reflecting on how important social media (and Facebook in particular) was in their everyday life. They also expressed how they wished there were other ways other than Facebook to complete their everyday tasks. All these findings inform the design of messages that should be sent to users in the event of an outage. In addition to Grandhi et al. [8]'s suggestion that we should provide more calming and reassuring messages than simple error messages to ease their anxiety, our study also reveals the importance of providing timely notifications and messages to users that address their concerns and explaining the potential solutions that the company is working on. In addition, it will also be helpful to provide enough backup infrastructure to help users conduct basic everyday communication tasks that might address some parts of their concerns when facing an outage. From the qualitative analysis of the quotes, we can see that there is a wide range of user concerns, ranging from losing connections with friends and family, the interruption of business operations, inability to sign into third-party websites, and the loss of entertainment. Future studies could further explore the top concerns or aspects of users' lives that were impacted by the outage, and practitioners can design backup infrastructure accordingly (e.g., alternative ways of signing in).

Furthermore, surprisingly, we found that a fair number of tweets pertained to the celebration of detox from Facebook. Several users

mentioned how it is important to experience the offline world and how peaceful they felt without the constant buzz from these social media platforms. This result not only implies the importance of designing alternative offline infrastructure that is less demanding for users, but also the possibility of incorporating a range of possible features on these inter-connected social media for users to actively undergo a social media detox if they would like to do so.

One the other hand, we also see an increasing trend of users searching for other social media alternatives as the outage continues into several hours. Moreover, the fact that #facebookdown trended number one on Twitter indicates that users would go to other social media to talk about the platform that is down, making the outage of one platform becoming major fodder for content in another platform. This kind of user migration calls for the importance of taking into account the social media ecosystems while understanding users' social media usage. For instance, Facebook (now Meta) has an official account on Twitter which could be used by the company to provide up-to-date news about the status of the outage and the anticipated time for resolution to its users in a timely fashion (the first tweet they sent out was 3 hours after the outage). This phenomenon implies that platforms could leverage the power of other social media platforms to communicate with users.

6 LIMITATION AND FUTURE WORK

In general, the current study reveals the major themes in users' reactions towards an unplanned social media outage. However, it is not free from limitations. First of all, we would like to address the nature of the dataset as all data came from Twitter users, who tend to be younger and more educated than Facebook users [17]. Moreover, the users who posted these tweets cannot be considered as representative of all Twitter users, as only English tweets were included and not all users participated in the discussion of Facebook outage. Yet, one of our major goals is to study how users express their reactions towards social media outage on alternative social media platforms, which is, by itself, a unique and interesting phenomenon. Future studies could expand our analysis to other social media platforms (e.g., Reddit, which tends to have more extensive discussions). Secondly, the current study adopted the LDA approach for topic modeling which might not be powerful to deal with shorter texts like tweets even though it is computationally inexpensive when working with a large dataset [21]. Future studies could deploy other methods to build topic models, such as biterm topic model (BTM) which is better at analyzing shorter text especially when the number of topics is large ($K > 40$) [20]. Finally, the current study relied solely on users' expression towards the outage which is not equal to users' perceptions and can be quite one-sided, as there are many users who chose to remain salient on social media when experiencing the outage. Furthermore, what we found in the current study might be particularly salient given the enhanced use of social media during the pandemic, which is characterized by lockdowns and remote working. Future empirical studies could build on our findings and use survey and qualitative methods (e.g. interviews or focus groups) to investigate users' perceptions and attitudes towards the outage more in depth.

ACKNOWLEDGMENTS

We thank Dr. Burt L. Monroe for his feedback on the paper and Ryan. Y. Wang for his assistance with data analysis.

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A APPENDICES

A.1 Data preprocessing

The quanteda package in R was used for data preprocessing and cleaning. All URLs, mentions (tags of other Twitter users), non-alphanumeric characters (i.e., emoticons), were removed and all words were lower-cased. Additionally, words that are too common based on both a self-generated list⁵ List of self-identified words removed: facebookdown, Facebookdown, facebookDown, FacebookDown, whatsappdown, FACEBOOKDOWN, facebook, Facebook, Down, down, Instagram, Instagram, Whatsapp, whatsapp, Whatsapdown, WhatsAppdown, WhatsAppDown, Instagram, Messenger, whatsapp, WhatsApp, insta, instagramdisabled, instagramdown, Instagramdown, InstagramDown, messengerdown, MessengerDown and the default stopwords list from the package were removed to avoid skewing the results. We created a document-term matrix (DTM) with unigrams (i.e., n-gram=1), in which a piece of text is decomposed into individual words. We retained the features that appeared in more than 50 documents or appeared in less than 80% of the documents, which helped remove 95.9% features (N=80,833), resulting in 3,450 features in the final DTM.

A.2 LDA model selection

Based on past literature regarding involuntary non-use [9], the number of topics should be at least three (positive personal impact, negative personal impact, societal impact), so we developed

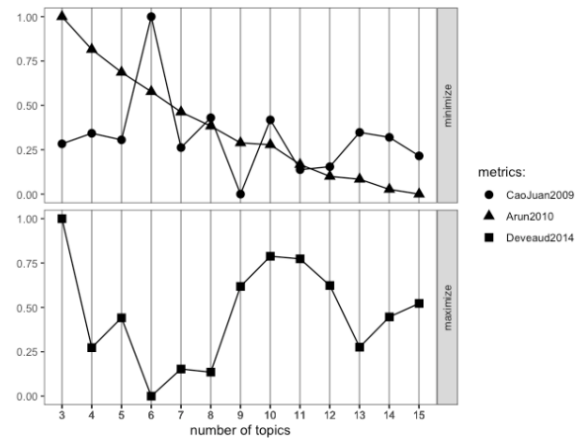


Figure 3: Results from ldatuning with the CaoJuan 2009, Arun 2010 and Deveaud 2014 matrix respectively (k=3 to 15)

⁵List of self-identified words removed: facebookdown, Facebookdown, facebookDown, FacebookDown, whatsappdown, FACEBOOKDOWN, facebook, Facebook, Down, down, Instagram, Instagram, Whatsapp, whatsapp, Whatsapdown, WhatsAppdown, WhatsAppDown, Instagram, Messenger, whatsapp, WhatsApp, insta, instagramdisabled, instagramdown, Instagramdown, InstagramDown, messengerdown, MessengerDown



Figure 4: Top 20 terms within each topic (based on the per-topic-per-word probabilities, beta) for the 9-topic LDA model

a number of models with k ranging from 3 to 15. We chose the maximize number of topics to be 15 to avoid getting topics that are too scattered to make theoretical sense. Using the package "ldatuning" in R, we chose the optimal number of topics based on three different matrices developed by past researchers [2, 6, 7], such that the CaoJian 2009 and Arun 2010 matrixes should be minimized and the Deveaud 2014 matrix should be maximized. Figure 3 shows the results from this model fitting process, and we can see when $k=9, 10$, or 11 , the models achieved optimal performances on the three matrixes.

We then trained the three models with 9, 10 or 11 topics respectively. The model that generated the most theoretically sound topics was retained, which was the model with 10 topics (See Figure 4 and Figure 5 below).

A.3 Results from the Word cloud of top 150 words in the corpus

Figure 6 below is the word cloud we generated based on all the common words users' used when discussing the Facebook outage incident.



Figure 5: Top 20 terms within each topic (based on the per-topic-per-word probabilities, beta) for the 11-topic LDA model

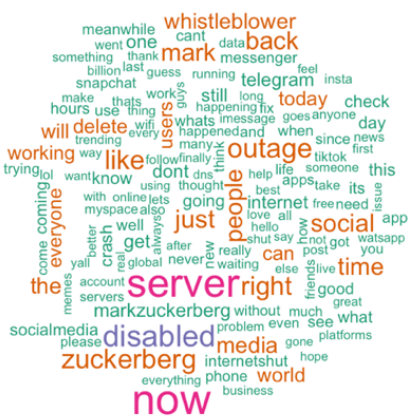


Figure 6: Word Cloud of the 150 most common words used in the tweets in general (after excluding the default stopwords in the quanteda package and the stopwords defined by us).